

Local AI on Jetson Nano

Jetson Nano 4GB
Ubuntu 18.04.6 / JetPack 4
Python 3.11



Setup

Run in a terminal on the Nano.

```
git clone https://github.com/nileshsarkar-ai/JetsonNano-Demo.git
```

```
cd JetsonNano-Demo
```

```
bash scripts/run_demo.sh --storage
```

```
bash scripts/run_demo.sh --check
```

```
bash scripts/run_demo.sh --setup-only
```

```
bash scripts/run_demo.sh
```

Text Conversation

Menu 3

SmolLM2-360M Instruct

Ask a question, then follow up using the same conversation.

Example prompt

“Explain why a boat floats. Then explain it to a ten-year-old.”

/reset clears the conversation. /quit returns to the menu.

Camera Object Detection

Menu 4

SSD-Mobilenet-v2

Place a cup, bottle or book in view.

The detector labels objects and draws their bounding boxes.

Move an object and observe the changing detections.

Camera-Guided Object Hunt

Menu 5

- 1 The language model proposes objects to find.
- 2 Arrange the objects in view and press Enter to scan.
- 3 Camera detections update the found and remaining lists.

The hunt shares the conversation model and object detector.

Models and Execution

Text

SmolLM2-360M Q4
258 MiB model file
CPU inference

Camera

SSD-Mobilenet-v2
TensorRT on the Nano GPU
Camera and text stages run sequentially

Menu Controls

- 1 Prepare models and dependencies
- 2 Check board status
- 3 / 4 / 5 Run a demonstration
- 6 Inspect storage usage
- 7 Open this presentation
- 0 Exit

Ctrl+C cancels a demo and returns to the menu.